



## 1 Module Parameters

|                              |                                                                    |
|------------------------------|--------------------------------------------------------------------|
| Qualification                | Performance parameter                                              |
|                              | Functional and technical specifications                            |
| Application area             | Single barrel / double barrel / thermal sight / night vision sight |
| Transmitting aperture        | Φ 06                                                               |
| Receiving aperture           | 13*8                                                               |
| Range (building)             | 600, 800, 1000, 1200, 2000m                                        |
| Range accuracy (building)    | <200m±0.6m, over 200 m, 0.5%                                       |
| Maximum measurement time     | ≤ 0.9S                                                             |
| Laser divergence angle       | 3mrad                                                              |
| Type of laser                | 905nm (laser)                                                      |
| Pulse energy (nJ)            | Single pulse: 1800nJ                                               |
| Service voltage              | +3.3V~+3.8V                                                        |
| Communication interface      | UART (TTL level)                                                   |
| Static merit                 | 100mW@3.3V                                                         |
| Whistlers (db)               | < 60db                                                             |
| Module weight                | About 10.5g                                                        |
| Classification of waterproof | IPX7                                                               |
| Impact resistance rating     | 1000G                                                              |
| External dimensions          | 27.5*25*12.8mm                                                     |
| Working temperature          | 0℃~50℃                                                             |
| Storage temperature          | 0℃~60℃                                                             |
| Baud rate                    | 115200 bps/9600 bps (default 9600 bps)                             |



Pin definition diagram for the RS-485 module:

- 1:GND
- 2:VCC
- 3:NC
- 4:Uart Tx
- 5:Uart Rx
- 6:Power ON (高电平有效)

| Baotou 1<br>(1 byte) | Baotou 2<br>(1byte) | Data length<br>(1 byte) | Address<br>code (1 by-<br>te) | Command<br>word (1 by-<br>te) | Data field<br>(1 byte) | Checksum<br>(1 byte) | Wrap 1<br>(1byte) | Wrap 2<br>(1byte) |
|----------------------|---------------------|-------------------------|-------------------------------|-------------------------------|------------------------|----------------------|-------------------|-------------------|
| AE                   | A7                  |                         |                               |                               |                        |                      | BC                | BE                |

| Baotou 1<br>(1byte) | Baotou 2<br>(1byte) | Data length<br>(1 byte) | Address<br>code (1 by-<br>te) | Command<br>word (1 by-<br>te) | Data field<br>(1 byte) | Checksum<br>(1 byte) | Wrap 1<br>(1byte) | Wrap 2<br>(1byte) |
|---------------------|---------------------|-------------------------|-------------------------------|-------------------------------|------------------------|----------------------|-------------------|-------------------|
| AE                  | A7                  | 05                      | 00                            | 0B                            | 00                     | 10                   | BC                | BE                |

| Baotou 1<br>(1byte) | Baotou 2<br>(1byte) | Data length<br>(1 byte) | Address<br>code (1 by-<br>te) | Command<br>word (1 by-<br>te) | Data field<br>(0 bytes) | Checksum<br>(1 byte) | Wrap 1<br>(1byte) | Wrap 2<br>(1byte) |
|---------------------|---------------------|-------------------------|-------------------------------|-------------------------------|-------------------------|----------------------|-------------------|-------------------|
| AE                  | A7                  | 04                      | 00                            | 05                            |                         | 09                   | BC                | BE                |

| Baotou 1<br>(1byte) | Baotou 2<br>(1byte) | Data length<br>(1 byte) | Address<br>code (1 by-<br>te) | Command<br>word (1 by-<br>te) | Data field<br>(19 bytes) | Checksum<br>(1 byte) | Wrap 1<br>(1byte) | Wrap 2<br>(1byte) |
|---------------------|---------------------|-------------------------|-------------------------------|-------------------------------|--------------------------|----------------------|-------------------|-------------------|
| AE                  | A7                  | 17                      | 00                            | 85                            | MMSG*                    |                      | BC                | BE                |

[illegible]

3.2.4 continuous measurement control  
Send command: AE A7 04 00 0E 12 BC BE:

| Table 3-4 Continuous Measurement Control: |                     |                         |                               |                          |                         |                      |                   |                   |
|-------------------------------------------|---------------------|-------------------------|-------------------------------|--------------------------|-------------------------|----------------------|-------------------|-------------------|
| Baotou 1<br>(1byte)                       | Baotou 2<br>(1byte) | Data length<br>(1 byte) | Address<br>code (1 by-<br>te) | Command<br>word (1 byte) | Data field<br>(0 bytes) | Checksum<br>(1 byte) | Wrap 1<br>(1byte) | Wrap 2<br>(1byte) |
| AE                                        | A7                  | 04                      | 00                            | 0E                       |                         | 12                   | BC                | BE                |

Upon receiving the command, respond with: AE A7 04 00 8E 92 BC  
BE. Initiate continuous measurement and respond upon successful  
measurement.

| Table 3-5 Continuous Measurement Response: |                     |                         |                               |                          |                          |                      |                   |                   |
|--------------------------------------------|---------------------|-------------------------|-------------------------------|--------------------------|--------------------------|----------------------|-------------------|-------------------|
| Baotou 1<br>(1byte)                        | Baotou 2<br>(1byte) | Data length<br>(1 byte) | Address<br>code (1 by-<br>te) | Command<br>word (1 byte) | Data field<br>(19 bytes) | Checksum<br>(1 byte) | Wrap 1<br>(1byte) | Wrap 2<br>(1byte) |
| AE                                         | A7                  | 17                      | 00                            | 85                       | MMSG*                    |                      | BC                | BE                |

\*MMSG is the returned measurement result, defined as follows:

| Elevation | Currently distance | Site height | Horizon range | Topographic | Azimuth | Horizontal angle | Span    | Velocity | Parasang |
|-----------|--------------------|-------------|---------------|-------------|---------|------------------|---------|----------|----------|
| (2byte)   | (2byte)            | (2byte)     | (2byte)       | (2byte)     | (2byte) | (2byte)          | (2byte) | (2byte)  | (1byte)  |

explain :  
The measured value occupies two bytes and is transmitted as a signed short type with the high 8 bits first and the low 8 bits last. Units: Angle: 0.1 degrees; Speed: 0.1 km/h; Distance: 0.1 meters (01),0.1 yards (02),0.1 feet (03). Note: The device supports only one linear distance measurement.  
Response command for measurement failure: AE A7 04 00 0E 12 BC BE

3.2.5 Stop continuous measurement  
Stop continuous ranging control command: AE A7 04 00 0F 13 BC BE

| Baotou 1<br>(1byte) | Baotou 2<br>(1byte) | Data length<br>(1 byte) | Address<br>code (1 by-<br>te) | Command<br>word (1 byte) | Data field<br>(0 bytes) | Checksum<br>(1 byte) | Wrap 1<br>(1byte) | Wrap 2<br>(1byte) |
|---------------------|---------------------|-------------------------|-------------------------------|--------------------------|-------------------------|----------------------|-------------------|-------------------|
| AE                  | A7                  | 04                      | 00                            | 0F                       |                         | 13                   | BC                | BE                |

Response to the command: AE A7 04 00 8F 93 BC BE

IV. Factors Affecting Range Measurement Capability, Response Speed and Velocity Measurement Accuracy

4.1 Target reflectivity: Generally, higher target reflectivity results in better ranging performance and faster response times. For instance, medium reflectivity targets can be measured up to 900 meters, high reflectivity targets can reach depths of at least 2000 meters, while low reflectivity targets may only be detectable up to 600 meters. (Diffuse reflection targets like water surfaces might be unmeasurable.)

4.2 Target shape: When the measured target's reflective surface area is too small or uneven, the ranging capability and ranging response speed will be correspondingly reduced;

4.3 Measurement Angle: When the laser beam is vertically directed at the target's reflective surface, the ranging capability and response speed are optimized. Conversely, both performance metrics will degrade under oblique angles. Note that using extreme measurement angles may compromise the ranging capability and response speed specified in this manual (under special conditions, the angle between black smooth planes must be no less than 60°).

4.4 Measurement Environment: Factors affecting ranging capability and response speed include solar radiation intensity, concentrations of water vapor and suspended particles in the air, as well as angles of deviation from sunlight exposure. Laser reflectivity and measurement results may vary depending on climatic conditions, environmental factors, the target's color, surface finish, size, actual shape, and other characteristics.

Measurements may be inaccurate or fail under the following conditions:

- ◆Snow, rain, or fog
- ◆Small or slender target
- ◆Target is black or dark
- ◆target with a graded surface
- ◆Target is moving or vibrating
- ◆When measuring the water surface
- ◆target measured through glass
- ◆Target is glass or mirror
- ◆The incident angle of the laser to the target surface is oblique
- ◆Sunlight incidence angle ≥30°
- ◆Visibility ≥10 km

4.5 Measuring reaction speed: The product uses an invisible laser beam to measure the time it takes for the laser beam to travel from the range telescope to the target and back.

Maximum measurement time: 0.9 seconds  
Change target object response time to <0.9S  
Note: Inconsistent object reflectivity and module stability in locking the target may affect the module response speed.

V. Precautions

- 5.1 Do not look directly at the laser beam when using this module.
- 5.2 Do not operate this module with the lens barrel or any other optical accessories to prevent additional eye injury.
- 5.3 Do not remove the module. Removal is not covered by warranty.
- 5.4 When handling, storing, or using this product, avoid humid environments. Exposure to condensation or frost may impair measurement accuracy and potentially damage the module.
- 5.5 Add sufficient cushioning material in the packaging box during transportation to prevent damage to the module.
- 5.6 Keep the module out of reach of children and avoid placing it on unstable high places to prevent damage from falls.
- 5.7 Avoid placing the module in direct sunlight, near UV-exposed areas, or close to heat sources to prevent unintended damage.
- 5.8 If the module surface fogging occurs due to sudden temperature changes, do not use the module.
- 5.9 If the lens is dirty, gently wipe it with a lens cloth. Do not use other items to avoid scratching the lens surface.
- 5.10 This module comes with a one-year warranty and lifetime support. For quality-related issues, replacement is free. For human-induced problems, repair and replacement costs will be charged based on the actual condition of the item.

6. The Range of This Ranging Module Is Defined Under the Following Conditions:

- 6.1 The target reflectivity is over 85%.
- 6.2 The target surface is perpendicular to the laser emission direction.
- 6.3 The weather was clear but not under direct sunlight.
- 6.4 Tip: When measuring distant targets, use a tripod to stabilize the module to reduce its vibration during measurement and achieve better results.
- 6.5 Unconventional testing environments and methods not described in the table require mutual agreement and collaboration with the client to resolve related issues. The results of unconventional testing shall not be used as a basis for product quality assessment until confirmed.



arbor



Highway sign



Building Wall